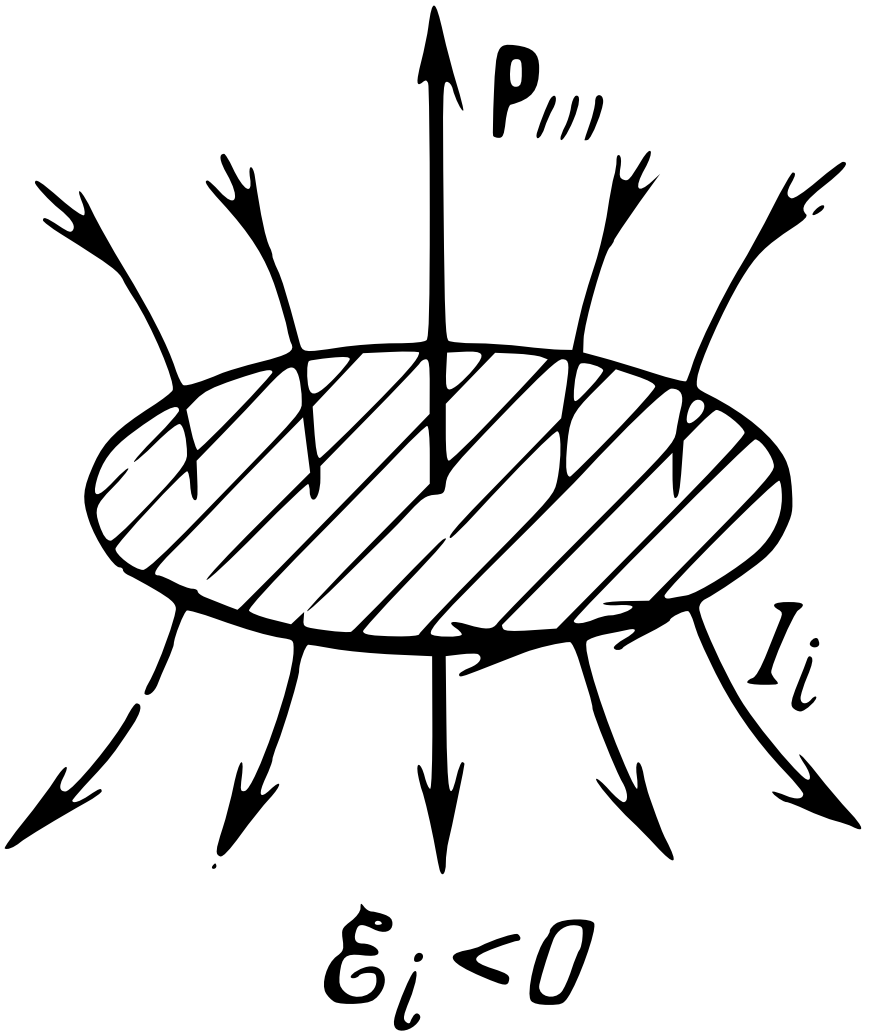
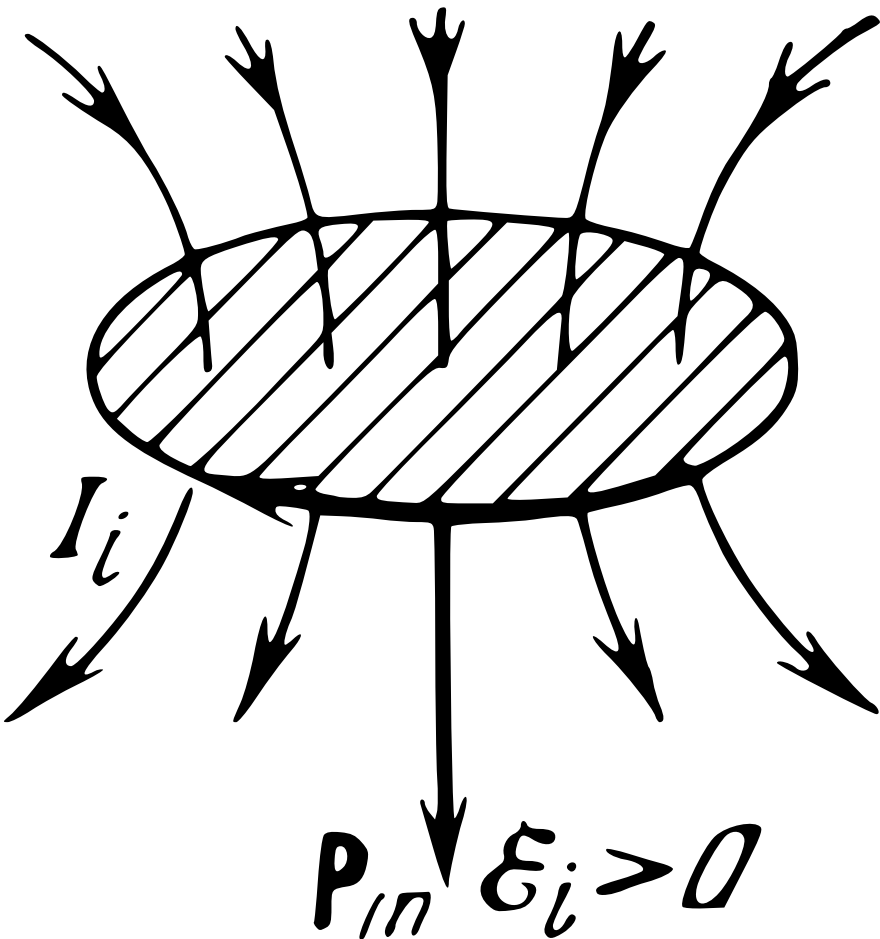
(a) $\frac{d\Phi_m}{dt} > 0$



(b) $\frac{d\Phi_{III}}{dt} < 0$